

SURAJ PERICHARLA

DATA ENGINEER

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SUMMARY

- Data Engineer with 5 years of experience in architecting and managing relational (MySQL, PostgreSQL, Oracle) and NoSQL (MongoDB) databases, proficient in ETL tools like Apache Airflow and AWS Glue.
- Proficient in big data tools (Apache Spark, Hadoop, Databricks) and cloud platforms (AWS, Azure) for large-scale data processing and machine learning.
- Skilled in Python, SQL, and data security, ensuring HIPAA, PCI DSS, and GDPR compliance with AWS KMS and Azure Key Vault.
- Experienced in real-time data streaming and analytics with Apache Kafka and AWS Kinesis, enabling quick decision-making from large-scale datasets.

SKILLS

Methodologies:	SDLC, Agile, Waterfall
Programming Language:	Python, SQL, R
Packages:	NumPy, Pandas, Matplotlib, SciPy, Scikit-learn, TensorFlow, Seaborn
Visualization Tools:	Tableau, Power BI, Advanced Excel (Pivot Tables, VLOOKUP)
Database:	MySQL, PostgreSQL, MongoDB, SQL Server
Data Engineering Concept:	Apache Spark, Apache Hadoop, Apache Kafka, Apache Beam, ETL/ELT, PySQL, PySpark
Cloud Platforms:	AWS (EC2, S3, Lambda, Glue, Athena, SNS, RDS, EMR), Azure (Databricks, Blob Storage, Load Balancer, Synapse Analytics, Data Lake Storage), Google Cloud Platform
Other Technical Skills:	Data Lake, SSIS, SSRS, SSAS, Docker, Kubernetes, Jenkins, Terraform, Informatica, Snowflake, Google Big Query, Machine Learning Algorithms, Natural Language Processing, Big Data, Git, GitHub, Advanced analytics, Statistical Methods, Data Mining, Data warehousing, Data transformation, Critical Thinking, Presentation Skills, Problem-solving

RELEVANT EXPERIENCE

Data Engineer

May 2023– Present

Citi Group

Texas, USA

- Applied scalable cloud solutions on AWS, increasing client adoption by 40% and reducing infrastructure costs by 30%.
- Streamlined data transfer between Hadoop and relational databases using Apache Sqoop, reducing import/export time by 50%.
- Optimized Apache Hive queries for large-scale data processing, reducing execution time by 40%.
- Optimized EC2 instances for Spark jobs, enhancing parallelism and caching, cutting job completion time by 50%.
- Automated provisioning of EC2, Lambda, RDS, and EMR resources using Terraform, reducing setup time by 75%.
- Executed CI/CD pipelines with Jenkins, cutting deployment time by 60% and tripling release frequency.
- Developed real-time ETL pipelines from data lake to Snowflake, reducing data latency by 70% and enabling near real-time analytics.
- Optimized AWS Glue jobs and dev endpoints, reducing ETL runtime by 40% and boosting resource utilization by 35%.
- Implemented Spark Structured Streaming for real-time data processing, integrating data lake, Snowflake, and Kafka, handling 1 million events per second with sub-second latency.

Data Engineer

August 2020 – July 2022

Accenture

India

- Created Snowflake data warehouses, boosting query performance by 40% through efficient staging and metadata management.
- Built and managed data pipelines with Apache Airflow, integrating diverse data sources such as CRM systems, web analytics, and social media platforms into a unified platform.
- Developed custom Python scripts for data cleansing and transformation, improving data accuracy and quality by 30% for analysis.
- Created a robust data governance framework using AWS Glue Data Catalog for metadata management and data lineage tracking.
- Implemented DVC (Data Version Control) to track changes in customer datasets and machine learning models over time.

Data Engineer Intern

July 2019 – March 2020

TCS

India

- Developed optimized SQL queries for data extraction and transformation in PostgreSQL, improving query performance by 40% and reducing processing time for millions of daily transactions.
- Deployed a Kafka and Confluent real-time streaming system, cutting fraud detection time by 40% and enabling real-time oversight.
- Leveraged Apache Spark with PySpark for large-scale data processing and applied machine-learning models for anomaly detection.
- Designed and managed a secure AWS S3 data lake with partitioning and access controls for regulatory compliance.
- Leveraged Amazon Redshift for data warehousing, creating efficient data models to query historical transaction data quickly.
- Enforced data quality checks using Great Expectations, reducing false positives in fraud detection by 25%.

EDUCATION

University of Texas at Arlington

Texas, USA

Master of Science in Computer Science (CGPA-3.7/4.0)

GITAM University

Visakhapatnam, India

Bachelor of Technology in Computer Science and Engineering (GPA-8.53/10.0)